

The Hall Effect Experiment

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Abstract—In this experiment, we studied the Hall effect in an n-doped Germanium sample and measured how the Hall voltage changes with magnetic field and current. By fitting the data, we obtained a Hall coefficient of $R_H = (7.38 \pm 1.49) \times 10^{-3} \text{ m}^3/\text{C}$ and found that the setup introduces a magnetic-field offset of about $C \approx 100 \text{ mT}$. From longitudinal voltage measurements, the resistance of the sample was determined to be $R = 44.96 \pm 0.03 \Omega$, which gives a resistivity of $(2.14 \pm 0.44) \times 10^{-2} \Omega \cdot \text{m}$ and a conductivity of $46.7 \pm 9.6 \text{ S/m}$. Using these values, the carrier density was calculated as $n = (8.46 \pm 1.71) \times 10^{20} \text{ m}^{-3}$, and the mobility as $\mu = 0.345 \pm 0.099 \text{ m}^2/(\text{V} \cdot \text{s})$. Overall, the results show that the Hall method provides reasonable transport parameters for the sample.

I. INTRODUCTION

HISTORY

The Hall effect was first discovered by Edwin Hall in 1879[1], while he was working on current carriers and magnetic fields when he was a Ph.D. student at John Hopkins University. He noticed that a perpendicular potential difference appears in a current carrying wire under the influence of a magnetic field. However, it was a shocking result at that time, since the Maxwell Laws do not predict this kind of potential difference. This is because the presence of the electron was not known, and the classical theory of electrons has not been proposed yet.

Between 1880 and 1900, experiments conducted on the Hall effect accelerated rapidly[2]. These experiments provided solid evidence on the fact that current carriers in metals have a negative sign. This founding was way before Thomson's discovery of the electron.

In 1897, J.J. Thomson discovered the electron[3], which then resulted in him being awarded the Nobel Prize in Physics "in recognition of the great merits of his theoretical and experimental investigations on the conduction of electricity by gasses." The foundation of a negatively charged particle completed the interpretation and validity of the Hall effect, which was then concluded to be the most likely current carrier in metals.

In 1900, the classical theory of electrons was proposed by Paul Drude (also called the Drude model)[4, 5]. The model was the application of the kinetic theory for electrons as current carriers in metals. The model then leads to more sophisticated models for metals, such as the Debye model and the Einstein solid model. After the Drude model, Ohm's Law and the Hall effect were understood theoretically in a well-defined manner.

In later years, after the discovery of semi-conductors, it was concluded that there exists more than one current carrier. The holes (with the absence of free electrons) were also thought of

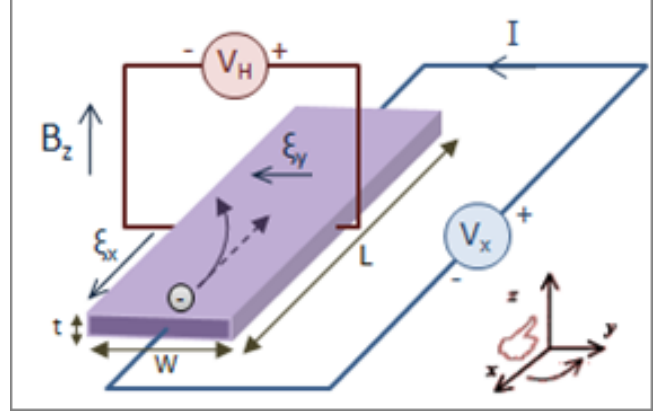


Fig. 1. This figure shows the Hall effect in a rectangular current carrier metal. The semi-conductor we used in the experiment have the exact same shape as in this figure. However, even working on this very basic model, provides us a very solid intuition about what is going on in the circuit.

as current carriers in metals. The n- and p-type semi-conductors were developed, and the Hall effect has become a widely used principle in solid state and semi-conductors physics.

THEORY

In the experiment, we used n-doped Germanium, so we effectively had only a single type of charge carrier, which has negative sign. So, we explain the theory of the Hall effect in single type charge carrier conductors. All the essentials of the Hall effect can be mainly derived by the use of the Lorentz Law, which is given in the following[6, 7].

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \quad (1)$$

In steady state, $\mathbf{F} = \mathbf{0}$ should be satisfied. Then the resulting equation is the following.

$$0 = E_y - v_x B_z \quad (2)$$

We also know the connection between the potential difference and the electric field. Thus, the electric field in terms of Hall potential is given by

$$w \xi_y = V_H \quad (3)$$

As shown in Fig.1, $E_y = -\xi_y$ and $v_x = -v_d$, where v_d is the drift velocity of the electrons. Rearranging these, we get the following formula for the Hall potential.

$$V_H = v_d B_z w \quad (4)$$

At this point, we need the Drude model to find the drift velocity. Since we are not concerned with the statistical derivation, we directly state the equation of motion for electrons in the metal as the Drude model proposed[7, 4, 5].

$$\frac{d}{dt}\langle \mathbf{p}(t) \rangle = q \left(\mathbf{E} + \frac{\langle \mathbf{p}(t) \rangle}{m} \times \mathbf{B} \right) - \frac{\langle \mathbf{p}(t) \rangle}{\tau} \quad (5)$$

where τ is called the **mean free time**, which means that the time an electron passes without colliding with another electron on average. Solving this equation in steady state and $\mathbf{B} = \mathbf{0}$, gives the following.

$$v_d = -\frac{e\tau}{m}E \quad (6)$$

where q is replaced by $-e$, which is the electron charge. We give a name to this prefactor.

$$\mu = \frac{e\tau}{m} \quad (7)$$

is called **electron mobility**. This is one of the things we need to calculate in the experiment. In a more compact matter, the drift velocity is expressed as $v_d = \mu E$.

Now, we try to find a connection between the electric field and the current density. To establish the relation between these, we examine the infinitesimal version of the Ohm's Law, which is given below[7].

$$\mathbf{J} = \sigma \mathbf{E} \quad (8)$$

where σ is called the **conductivity**. It is also equal to $\sigma = 1/\rho$, where ρ is called the **resistivity**. In Fig.1, we can write the current density as follows.

$$J_x = -nev_d \quad (9)$$

From Eq.9, we deduce the following two.

$$J = \frac{ne^2\tau}{m}E \quad (10)$$

$$\sigma = ne\mu \quad (11)$$

The configuration given in Fig.1 is easy to determine the relationship between the current I and the current density J . It is given by the following.

$$I_x = J_x wt \quad (12)$$

The definition of the Hall coefficient is as follows[6, 7].

$$\mathbf{E}_H = -R_H(\mathbf{J}_c \times \mathbf{B}) \quad (13)$$

Using Eq.13, we can finally find the following relations[8].

$$R_H = \frac{E_H}{J_x B_z} = \frac{V_H t}{I_x B_z} \quad (14)$$

$$V_H = \frac{R_H}{t} I B \equiv \alpha I B \quad (15)$$

In the analysis part, we will mainly use Eq.15. However, we can also find the explicit form of the Hall coefficient. Replacing

the unknowns in Eq.14 by Eq.12, Eq.9 and Eq.4, we can find the following expression for the Hall coefficient[6, 7].

$$R_H = -\frac{1}{ne} \quad (16)$$

Since the semi-conductor we used in the experiment is n-type conductor, we derived formula regarding this fact. However, in the most general case, there may be present a different type of charge carriers in the conductor. In this case, we need to modify the derivation procedure. Nevertheless, in the case where mixed-type charge carriers exist, the Hall coefficient is given by the following formula[6, 7].

$$R_H = \frac{p - nb^2}{e(p + nb)^2} \quad (17)$$

where $b = \mu_e/\mu_h$ is the ratio of the mobilities.

II. THE METHOD

In this section, we give the procedure to conduct the experiment. After successfully gathering the data sets, in the analysis part, we will show how to calculate the quantities we want to find[8].

- 1) First, make sure that the power supply unit is properly connected to both the electromagnet and the Hall probe. Be careful when connecting it to the Hall probe. It is depicted on the Hall probe how to connect the main current carrying cable. After properly connecting them, turn on the power supply unit. Throughout all the measurement parts, do not set the current beyond 100 mA and magnetic field beyond 200 mT.
- 2) Place both the magnetic field probe (denoted 1 in Fig.2) and the Hall probe (denoted as 3 in Fig.2) between the ends of the electromagnet (denoted 2 in Fig.2). The height and horizontal positions of both probes can be calibrated using their stands. Set the off-set to zero in the teslameter (denoted 6 in Fig.2).
- 3) Connect the multimeter to the Hall probe in the perpendicular orientation with respect to the main current passing through the circuit. On the probe, it is shown how to connect the multimeter to measure the Hall potential.
- 4) We arranged the setup properly, now, we can start to take the data sets. To take the first data set, set current to 50 mA and change the observed magnetic field (via increasing the applied voltage to the electromagnet) between 0 and 200 mT. Take 10 different measurements of Hall voltage across the Hall probe and create a data set containing 10 data points of Hall voltage vs. the magnetic field.
- 5) Set the magnitude of the observed magnetic field to zero and keep it fixed. Change the magnitude of the current between 0 and 100 mA, and again take 10 measurements of the Hall potential.
- 6) In the third measurement, we need to do a little rearrangement. We need to connect the multimeter the same way as the current passes. We want to measure the applied voltage V_{cc} in this part. After properly reconnecting the multimeter, we take the data as the current changes between 0 and 100 mA. In this measurement, the observed

magnetic field is irrelevant, so we can set it to any value. Overall, again make 10 measurements.

III. THE EXPERIMENTAL SETUP

In this section, we post the scheme of the experimental setup and list all the instruments that were used in the experiment with their names and properties[8].

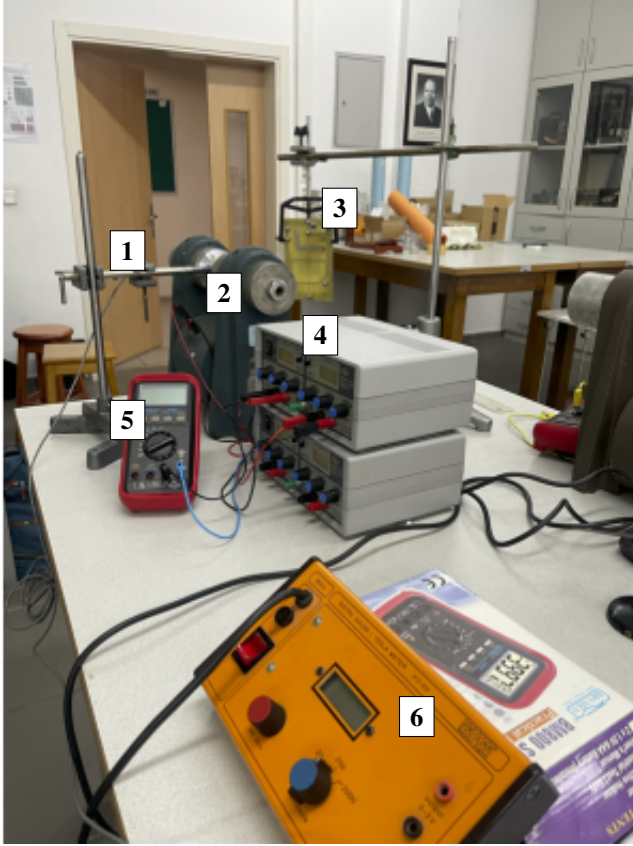


Fig. 2. This figure illustrates the setup and the experimental setups we used conducting the Hall effect experiment.

- 1) **Magnetic Field Probe:** It is used to determine the value of the magnetic field present between the ends of the electromagnet. This is connected to the teslameter shown as number 6 in Fig.2. Ironically, working mechanism of the probe relies on the Hall effect, i.e. inside the probe, there is a mini Hall circuit to determine the magnitude of the magnetic field.
- 2) **Electromagnet:** It is a very powerful electromagnet. As its name explains, it is used to create a constant magnetic field, and the magnitude of the magnetic field it creates is controlled by the voltage applied.
- 3) **Hall Probe:** This circuit is the heart of the experiment. It contains n-doped Germanium and we conduct the entire experiment on this circuit. It is very similar to one in the magnetic field probe.
- 4) **Power Supply, Voltmeter, and Ammeter:** It is used mainly as a power supply, but it also has a built-in voltmeter and ammeter. We supply a potential to both the electromagnet and Hall probe.

- 5) **Multimeter:** It has a few different modes to measure voltage, current, etc. We used it to measure the Hall voltage appearing across the Hall probe.
- 6) **Teslameter:** It is used to read the magnitude of the magnetic field measured by the probe (denoted 1). It is hard to directly measure the magnitude of the magnetic field. Instead, we measure the change in the magnetic field. Thus, the teslameter allows us to set the off-set to zero. We also try to calculate this off-set in the analysis part. It is crucial to distinguish between the magnetic field and the *observed* magnetic field.

IV. THE DATA

In this section, we share our data sets. In the experiment, as discussed in the Methods section, we take 3 different data sets, each consisting of 10 data points. We give them as tables as follows.

TABLE I
 B_0 vs. V_H VALUES AT FIXED CURRENT $I = 49.8 \pm 0.1$ mA

	B_0 (mT) ($\pm 5\%$ uncertainty)	V_H (mV)
1	20.3 ± 1.0	44.1 ± 0.1
2	40.2 ± 2.0	51.5 ± 0.1
3	60.3 ± 3.0	58.9 ± 0.1
4	80.3 ± 4.0	66.3 ± 0.1
5	99.9 ± 5.0	73.5 ± 0.1
6	119.9 ± 6.0	80.8 ± 0.1
7	140.4 ± 7.0	88.2 ± 0.1
8	160.6 ± 8.0	95.5 ± 0.1
9	175.7 ± 8.8	101.0 ± 0.1
10	190.4 ± 9.5	106.3 ± 0.1

TABLE II
 I vs. V_H VALUES AT FIXED OBSERVED MAGNETIC FIELD $B_0 = 0$ T

	I (mA)	V_H (mV)
1	5.17 ± 0.01	3.8 ± 0.1
2	15.14 ± 0.01	11.0 ± 0.1
3	24.76 ± 0.01	18.1 ± 0.1
4	35.18 ± 0.01	25.8 ± 0.1
5	45.4 ± 0.1	33.4 ± 0.1
6	55.4 ± 0.1	40.7 ± 0.1
7	64.7 ± 0.1	47.8 ± 0.1
8	75.0 ± 0.1	55.8 ± 0.1
9	85.5 ± 0.1	64.1 ± 0.1
10	95.2 ± 0.1	72.1 ± 0.1

TABLE III

I vs. V_{cc} VALUES. THE MAGNITUDE OF THE OBSERVED MAGNETIC FIELD IS IRRELEVANT IN THIS CASE. HOWEVER, WE SET $B_0 = 0$ T.

	I (mA)	V_{cc} (mV)
1	3.85 ± 0.01	171.4 ± 0.1
2	12.45 ± 0.01	551 ± 1
3	17.26 ± 0.01	764 ± 1
4	25.10 ± 0.01	1112 ± 1
5	31.56 ± 0.01	1401 ± 1
6	40.9 ± 0.1	1816 ± 1
7	49.8 ± 0.1	2219 ± 1
8	59.4 ± 0.1	2664 ± 1
9	80.2 ± 0.1	3633 ± 1
10	93.3 ± 0.1	4300 ± 1

V. THE ANALYSIS

In this section, we perform the analysis of the experiment. In this experiment, we may find the experimental values of more than one quantity. The primary aim of the experiment -hence, the most important variable- is to find the experimental value of the Hall coefficient R_H .

In Eq.15, we found the relation between the Hall potential and the current and the magnetic field that we applied. However, as discussed, the magnetic field may have an off-set. So, we modify the magnetic field as follows.

$$B = B_0 + C \quad (18)$$

where B_0 is called the **observed magnetic field**. Then, Eq.15 becomes[8]

$$V_H = \alpha I B_0 + \alpha I C. \quad (19)$$

In the first part of the analysis, we use the data set given in Tab.I where the current I was kept fixed. Applying a linear fit to this graph, we can find its slope, which gives us information about the value of α .

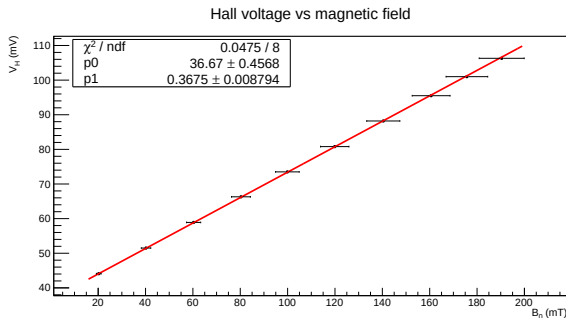


Fig. 3. This figure shows the V_H vs. B_0 graph with data points in Tab.I regarding their uncertainties. It seems that the uncertainty in the y-axis is not included, however, it is not the case. This is because the uncertainty in V_H is very small in comparison to its value. The line fit is applied to the graph and both the slope and the intersection values can be read from the box on the top left of the graph.

In Fig.3, we see that the values of the slope and the interception point are $p_1 = 0.3675 \pm 0.0088$ V/T and $p_0 =$

36.67 ± 0.46 mV, respectively. Since the uncertainty in B_0 is 5%, the error bars on the x-axis become very large, and this results in the failure of the χ^2/ndf test. The line fit assumes the form of V_H as follows[9].

$$V_H = p_1 B_0 + p_0. \quad (20)$$

From this, we can calculate R_H and C .

$$R_H = \frac{p_1 t}{I} \quad (21)$$

$$C = \frac{p_0}{p_1} \quad (22)$$

Their propagation of errors is given by the following[9].

$$\sigma_{R_H} = \sqrt{\left(\frac{t}{I}\right)^2 \sigma_{p_1}^2 + \left(\frac{p_1}{I}\right)^2 \sigma_t^2 + \left(\frac{p_1 t}{I^2}\right)^2 \sigma_I^2} \quad (23)$$

$$\sigma_C = \sqrt{\left(\frac{1}{p_1}\right)^2 \sigma_{p_0}^2 + \left(\frac{p_0}{p_1^2}\right)^2 \sigma_{p_1}^2} \quad (24)$$

We will take the value of thickness $t = 1.0 \pm 0.2$ mm and the value of current $I = 49.8 \pm 0.1$ mA as given in Tab.I. Then R_H and C are found as follows.

$$R_H = (7.38 \pm 1.49) \times 10^{-3} \text{ m}^3/\text{C} \quad (25)$$

$$C = 99.75 \pm 2.69 \text{ mT} \quad (26)$$

Using the data set given in Tab.II, we can also calculate value of the off-set C with different approach. It allows us to check whether the theoretical formulation $V_H = \alpha I (B_0 + C)$ correct.

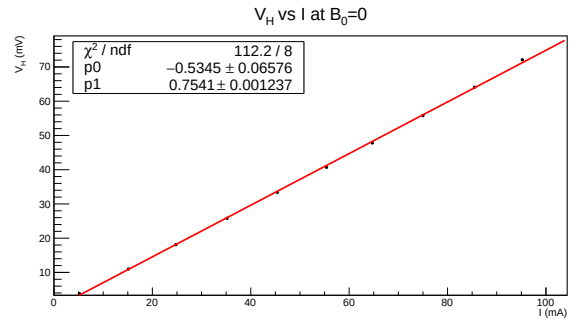


Fig. 4. This figure illustrates the relation between the applied current and the Hall voltage. Note that we measured the values at the constant observed magnetic field $B_0 = 0$ T. As in Fig.3, error bars are included. However, as it can be seen in Tab.II, uncertainty for both the current and the voltage values are very low compared to their values.

The line fit in Fig.4 gives us the following formula[9].

$$V_H = p_1 I + p_0 \quad (27)$$

where $p_1 = R_H C / t$ and $p_0 = V_0$. Since $B_0 = 0$, p_0 is not related to B_0 in this approach. Therefore, it gives us information about how noisy the system is (i.e. instrumental errors) and also how well the experimental setup was designed. In an ideal scenario, p_0 should be zero. Although in our setup

it is non-zero, it attains a relatively small value, which is more than enough for this kind of experiment. Using R_H value found in the first part, we calculate the value of C for the second graph. The formula for C with its propagation of error is given by the following[9].

$$C = \frac{p_1 t}{R_H} \quad (28)$$

$$\sigma_C = \sqrt{\left(\frac{t}{R_H}\right)^2 \sigma_{p_1}^2 + \left(\frac{p_1}{R_H}\right)^2 \sigma_t^2 + \left(\frac{p_1 t}{R_H^2}\right)^2 \sigma_{R_H}^2} \quad (29)$$

Thus, it is found as

$$C = 102 \pm 29 \text{ mT}. \quad (30)$$

The error is very high because the errors of R_H and t are relatively large. Nevertheless, the two values of C we found in the first and second parts are in agreement. This result allows us to use the theoretical approach we have adopted and it also shows that the statistical errors in the analysis are considerably small.

Now, we move on to the third part. In this part, we will be dealing with the longitudinal transition of the circuit. That is, we will try to find the resistivity (conductivity) of the circuit. These results will allow us to compute the electron mobility and the drift velocity using the previously found value of the Hall coefficient. Moreover, at the end of this section, we will also calculate the density of the charge carriers. In Fig.5, we can observe the relationship between the longitudinal voltage V_{cc} and the current I . Thus, the slope of the graph should give us the longitudinal resistance of the circuit.

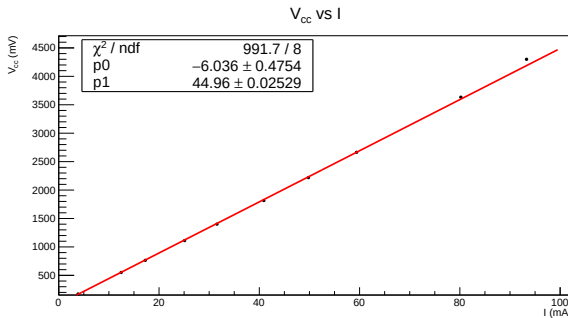


Fig. 5. This figure shows the data points given in Tab.III. The data points were plotted with their given uncertainties, however, comparing the scale of the graph, the uncertainties would be very low so that they can not be seen. The slope of the graph give us the resistance of the circuit. Since the errors are so low, the wellness of the fit and also the error of slope are quite poor.

Observing Fig.5, we conclude that the value of the longitudinal resistance is as follows.

$$R_{\text{long.}} = 44.96 \pm 0.03 \Omega \quad (31)$$

The geometry of the circuit is fortunately rectangular; thus it is easy to connect the microscopic and macroscopic Ohm's Laws to each other. Starting with the macroscopic version,

$$V = RI \quad (32)$$

$$\frac{V}{l} = \frac{R}{l} \frac{I}{wt} \quad (33)$$

$$E = \left(\frac{wt}{l} R\right) J \quad (34)$$

$$\Rightarrow \rho = R \frac{A}{l} \quad (35)$$

Here, ρ is called resistivity and its unit is $\Omega \cdot \text{m}$. We previously declared the numerical value of thickness, which is $t = 1.0 \pm 0.2 \text{ mm}$. The width and length values are $w = 1.00 \pm 0.05 \text{ cm}$ and $l = 2.10 \pm 0.05 \text{ cm}$, respectively. The error propagation for resistivity is given by the following[9].

$$\frac{\sigma_\rho^2}{\rho^2} = \frac{\sigma_R^2}{R^2} + \frac{\sigma_w^2}{w^2} + \frac{\sigma_t^2}{t^2} + \frac{\sigma_l^2}{l^2} \quad (36)$$

Regarding the equations above, we find resistivity as

$$\rho = (2.14 \pm 0.44) \times 10^{-2} \Omega \cdot \text{m}. \quad (37)$$

The relationship between resistivity and conductivity is given by $\rho = 1/\sigma$. Using this, we can also calculate the value of the conductivity.

$$\sigma = (46.7 \pm 9.6) \text{ S/m} \quad (38)$$

where the unit S is called Siemens and is an equivalent of $1/\Omega$. Therefore, using Eq.11, we can write the electron mobility in terms of resistivity and the Hall coefficient.

$$\rho = \frac{1}{ne\mu} \quad (39)$$

$$\rho = \frac{|R_H|}{\mu} \quad (40)$$

$$\mu = \frac{|R_H|}{\rho} \quad (41)$$

Using the equation above, we find the electron mobility as

$$\mu = 0.344 \pm 0.099 \text{ m}^2/(\text{V} \cdot \text{s}) \quad (42)$$

The drift velocity v_d can be calculated using the electron mobility. However, the drift velocity is not a distinct value. Instead, it is proportional to the magnitude of the electric field. We just want to avoid too many calculations regarding all the different values of current or voltage that we measured in the experiment. In addition, there is still a remaining, yet important, quantity that we can compute. It is the charge carrier density and is given by the following equation.

$$n = \frac{1}{e|R_H|} \quad (43)$$

The true value of the electron charge is $e = 1.602 \times 10^{-19} \text{ C}$. Then, the charge carrier density is given by

$$n = (8.46 \pm 1.71) \times 10^{20} \text{ m}^{-3} \quad (44)$$

VI. THE RESULT

In this section, we declare our finding through the analysis. In this experiment, there were many quantities that could be calculated. These can be found in the following.

- $R_H = -(7.38 \pm 1.49) \times 10^{-3} \text{ m}^3/\text{C}$
- $C = 99.75 \pm 2.69 \text{ mT}$
- $R = 44.96 \pm 0.03 \Omega$
- $\rho = (2.14 \pm 0.44) \times 10^{-2} \Omega \cdot \text{m}$
- $\sigma = 46.7 \pm 9.6 \text{ S/m}$
- $\mu = 0.344 \pm 0.099 \text{ m}^2/(\text{V} \cdot \text{s})$
- $n = (8.46 \pm 1.71) \times 10^{20} \text{ m}^{-3}$

VII. THE CONCLUSION

There are a few things we want to discuss about the experiment. Firstly, in the analysis part, the Hall coefficient was calculated with a wrong sign. This was neither a computational nor a statistical mistake. This was due to the wrongly placed ends of the multimeter during the measurement in the experiment. However, since we know that the semi-conductor we are dealing with is an n-doped, we put a minus sign in front of the value of R_H in the result section.

The other thing we want to discuss is the wellness of the fits and error bars in the graphs. In graphs except one including the magnetic field data, the error bars are so low that they can not be seen. However, we ensure that we define the error bars in the code. This is because we take the uncertainty of the data points as much as the significant figures that the measurement device reads. Apparently, it was not a good approach; hence we needed to consider the uncertainties of the measurement devices as a percentile.

Another crucial remark is that the uncertainties of length values of the circuit are relatively large, especially the uncertainty of the value of thickness. This is why the error of the Hall coefficient is found to be that big.

The last thing is that there is no nominal value by which we can compare our findings. This is because the nature of the Hall effect experiment is a bit unique. We used n-doped Germanium as the semi-conductor. However, we do not have any information about the amount of doping. As it changes; the Hall coefficient, electron mobility, resistivity (and naturally conductivity) change. Thus, it is hard to come to a conclusion about the results. However, in Germanium, the electron mobility should be on the order of $4,000 \text{ cm}^2/(\text{V} \cdot \text{s})$ [10]. This is exactly what we found.

REFERENCES

- [1] Edwin H. Hall. “On a new action of the magnet on electric currents”. In: *American Journal of Mathematics* 2.3 (1879), pp. 287–292. DOI: 10.2307/2369245.
- [2] J. J. Thomson. *Electricity and Matter*. New York: Charles Scribner’s Sons, 1903.

- [3] J. J. Thomson. “Cathode Rays”. In: *Philosophical Magazine* 44.269 (1897), pp. 293–316. DOI: 10.1080/14786449708621070.
- [4] Paul Drude. “Zur Elektronentheorie der Metalle”. In: *Annalen der Physik* 306.3 (1900), pp. 566–613. DOI: 10.1002/andp.19003060312.
- [5] Paul Drude. “Zur Elektronentheorie der Metalle”. In: *Annalen der Physik* 308.11 (1900), pp. 369–402. DOI: 10.1002/andp.19003081102.
- [6] Neil W. Ashcroft and N. David Mermin. *Solid State Physics*. New York: Holt, Rinehart and Winston, 1976.
- [7] Charles Kittel. *Introduction to Solid State Physics*. 8th ed. Hoboken, NJ: Wiley, 2004.
- [8] E. Gülmez. *Advanced Physics Experiments*. Boğaziçi University Publications, 1999.
- [9] E. Gülmez. *Basic Data Analysis for Experiments in the Physical Sciences*. Boğaziçi University, 1997.
- [10] Ioffe Institute. *Semiconductor Material Parameters Database*. <http://www.ioffe.ru/SVA/NSM/>. Accessed 2025.

VIII. APPENDIX

```
#include <TGraphErrors.h>
#include <TF1.h>
#include <TCanvas.h>
#include <TStyle.h>

void vcc_vs_I_linefit()
{
    gStyle->SetOptFit(1);

    const int N = 10;

    double x[N] = { 3.85, 12.45, 17.26, 25.10,
                    31.56, 40.9, 49.8, 59.4, 80.2, 93.3 };

    double ex[N] = { 0.01, 0.01, 0.01, 0.01,
                    0.01, 0.1, 0.1, 0.1, 0.1, 0.1 };

    double y[N] = { 171.4, 551, 764, 1112, 1401,
                    1816, 2219, 2664, 3633, 4300 };

    double ey[N] = { 0.1, 1, 1, 1, 1,
                    1, 1, 1, 1, 1 };

    TCanvas* c1 = new TCanvas("c1", "I vs V_{cc}", 800, 600);

    TGraphErrors* gr = new TGraphErrors(N, x, y, ex, ey);
    gr->SetTitle("V_{cc} vs I; I (mA); V_{cc} (mV)");
    gr->SetMarkerStyle(20);
    gr->SetMarkerSize(1.0);

    gr->Draw("AP");
```

```

TF1* f1 = new TF1("f1", "pol1", 0, 100);
gr->Fit(f1, "RE");

c1->Update();
}

#include <TGraphErrors.h>
#include <TF1.h>
#include <TCanvas.h>
#include <TStyle.h>

void vh_vs_b0_linefit()
{
    gStyle->SetOptFit(1);

    const int N = 10;

    double x[N] = { 20.3, 40.2, 60.3, 80.3,
                    99.9,
                    119.9, 140.4, 160.6, 175.7,
                    190.4 };
    double ex[N] = { 1.0, 2.0, 3.0, 4.0, 5.0,
                    6.0, 7.0, 8.0, 8.8, 9.5 };

    double y[N] = { 44.1, 51.5, 58.9, 66.3,
                    73.5,
                    80.8, 88.2, 95.5, 101.0, 106.3
                    };
    double ey[N] = { 0.1, 0.1, 0.1, 0.1, 0.1,
                    0.1, 0.1, 0.1, 0.1, 0.1 };

    TCanvas* c1 = new TCanvas("c1", "B vs V_H",
                              800, 600);

    TGraphErrors* gr = new TGraphErrors(N, x,
                                         y, ex, ey);
    gr->SetTitle("Hall voltage vs magnetic
                field;B_{0} (mT);V_{H} (mV)");
    gr->SetMarkerStyle(20);
    gr->SetMarkerSize(1.0);

    gr->Draw("AP");

    TF1* f1 = new TF1("f1", "pol1", 15.0,
                      200.0);
    gr->Fit(f1, "RE");

    c1->Update();
}

```

```

#include <TGraphErrors.h>
#include <TF1.h>
#include <TCanvas.h>
#include <TStyle.h>

void vh_vs_I_linefit()
{
    gStyle->SetOptFit(1);

    const int N = 10;

    double x[N] = { 5.17, 15.14, 24.76, 35.18,
                    45.4,

```

```

                    55.4, 64.7, 75.0, 85.5, 95.2 };

    double ex[N] = { 0.01, 0.01, 0.01, 0.01,
                    0.1,
                    0.1, 0.1, 0.1, 0.1, 0.1 };

    double y[N] = { 3.8, 11.0, 18.1, 25.8, 33.4,
                    40.7, 47.8, 55.8, 64.1, 72.1 };

    double ey[N] = { 0.1, 0.1, 0.1, 0.1, 0.1,
                    0.1, 0.1, 0.1, 0.1, 0.1 };

    TCanvas* c1 = new TCanvas("c1", "I vs V_H
                              at B0 = 0", 800, 600);

    TGraphErrors* gr = new TGraphErrors(N, x,
                                         y, ex, ey);
    gr->SetTitle("V_{H} vs I at B_{0}=0; I
                (mA); V_{H} (mV)");
    gr->SetMarkerStyle(20);
    gr->SetMarkerSize(1.0);

    gr->Draw("AP");

    TF1* f1 = new TF1("f1", "pol1", 0, 110);
    gr->Fit(f1, "RE");

    c1->Update();
}

```

The general formula for error propagation is given below. Define a function

$$y = f(x_1, \dots, x_n). \quad (45)$$

Then the error σ_y of y is given as[9]

$$\sigma_y^2 = \sum_k^n \left(\frac{\partial f}{\partial x_k} \right)^2 \sigma_{x_k}^2 + \sum_{l < k}^m \left(\frac{\partial f}{\partial x_k} \right) \left(\frac{\partial f}{\partial x_l} \right) \sigma_{kl}^2 \quad (46)$$

where σ_{kl} represents the relative error of the variable x_k to x_l . Also, $\sigma_{kl} = \sigma_{lk}$.

If the variables do not depend on each other (relatively free or $\sigma_{kl} = 0$ for all k and l values), equation 46 becomes[9]

$$\sigma_y^2 = \sum_k^n \left(\frac{\partial f}{\partial x_k} \right)^2 \sigma_{x_k}^2. \quad (47)$$